



CARIBBEAN EXAMINATIONS COUNCIL

ADVANCED PROFICIENCY EXAMINATION

PHYSICS

UNIT 2 – Paper 02

2 hours 30 minutes

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. This paper consists of SIX questions in two sections. Answer ALL questions.
2. For Section A, write your answers in the spaces provided in this booklet.
3. For Section B, write your answers in the spaces provided at the end of each question in this booklet.
4. Diagrams may be drawn in pencil but **all writing** must be in **ink**.
5. All working **MUST** be **CLEARLY** shown.
6. The use of silent, non-programmable calculators is permitted, but candidates should note that the use of an inappropriate number of figures in answers will be penalized.

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C.E.S.E.C.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

LIST OF PHYSICAL CONSTANTS

Speed of light in free space	c	=	$3.00 \times 10^8 \text{ m s}^{-1}$
Permeability of free space	μ_0	=	$4\pi \times 10^{-7} \text{ H m}^{-1}$
Permittivity of free space	ϵ_0	=	$8.85 \times 10^{-12} \text{ F m}^{-1}$
			$\frac{1}{4\pi\epsilon_0} = 9.0 \times 10^9 \text{ m F}^{-1}$
Elementary charge	e	=	$1.60 \times 10^{-19} \text{ C}$
Planck's constant	h	=	$6.63 \times 10^{-34} \text{ J s}$
Unified atomic mass constant (Energy equivalence:	u $1u$	= =	$1.66 \times 10^{-27} \text{ kg}$ 931 MeV/c^2
Rest mass of electron	m_e	=	$9.11 \times 10^{-31} \text{ kg}$
Rest mass of proton	m_p	=	$1.67 \times 10^{-27} \text{ kg}$
Acceleration due to gravity	g	=	9.81 m s^{-2}
1 Atmosphere	Atm	=	$1.00 \times 10^5 \text{ N m}^{-2}$
Avogadro's constant	N_A	=	$6.02 \times 10^{23} \text{ per mole}$

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SECTION A

Answer ALL questions in this section.

You MUST write your answers in the spaces provided in this booklet.

1. (a) Figure 1 shows a current balance used to determine the field between the poles of a permanent magnet. A current from the power supply flows through the pivots P, P' to the horizontal wire, AB, which experiences a downward force, F . This force is balanced by the torque caused by the weight, mg , of a small movable "slider".

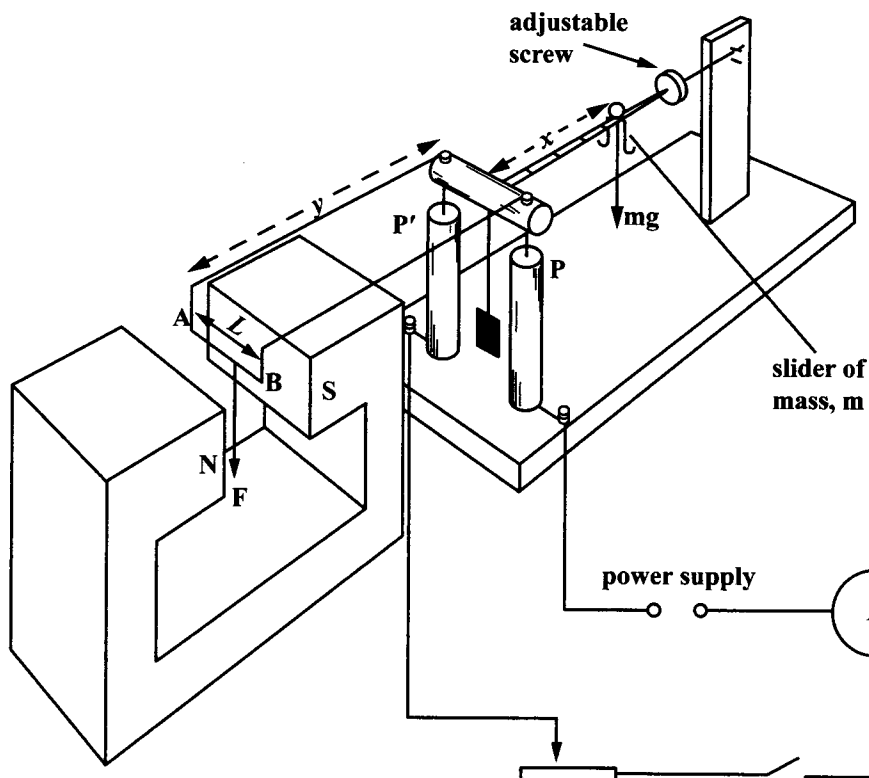
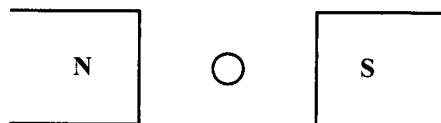


Figure 1

- (i) Mark on Figure 1 the direction of the current through the portion of the wire AB of the circuit and state the name of the rule used to determine the direction of the flow of current.
- (ii) Show, on the diagram below, the field lines between the poles of the magnet when the current is flowing in AB.



[3 marks]

- (b) (i) Show that at equilibrium, the current through the circuit is given by

$$I = \frac{mgx}{BLy}, \text{ where } L \text{ is the length of wire, AB.}$$

- (ii) Describe how the apparatus would be used to obtain a set of readings for I and x , clearly stating how the results are taken.

[6 marks]

- (c) Figure 2, on page 5, shows a graph of the data obtained for the current I/A versus position x/m .

- (i) Draw the best straight line through the data.

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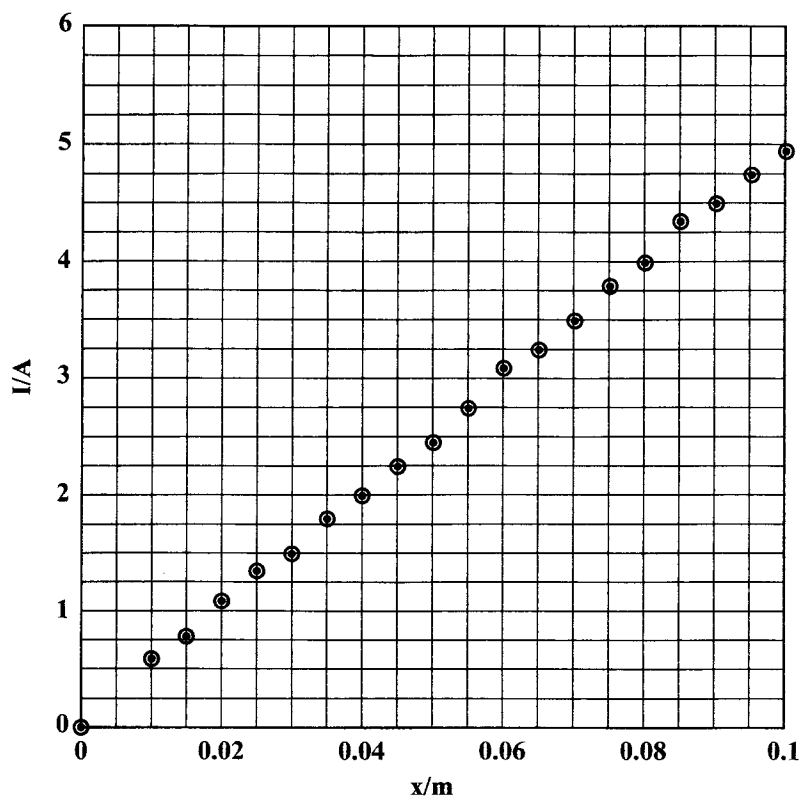


Figure 2

(ii) Determine the gradient of the best line drawn.

(iii) Given that $l = 2.0$ cm, $L = 5.0$ cm, $m = 0.50$ g, calculate the value of the magnetic flux density, B , for the magnet.

[6 marks]

Total 15 marks

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2. This question refers to the plotting of the transfer characteristic of a non-inverting amplifier.

- (a) (i) Add a resistor to the op-amp circuit below [Figure 3] so that it becomes a non-inverting amplifier with a theoretical gain of +10. Label it with the value of the required resistance.

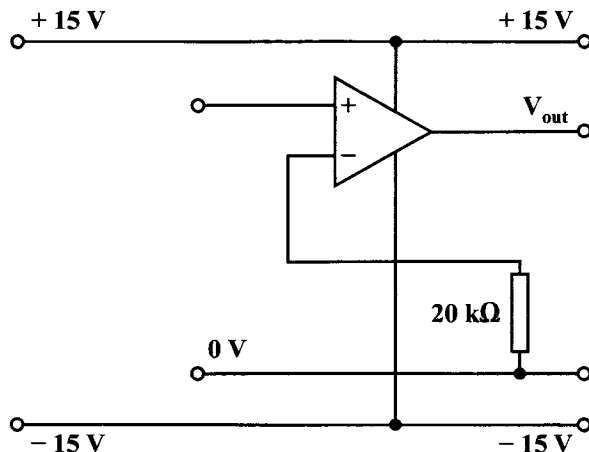


Figure 3

- (ii) Show on Figure 3 how meters and a potential divider would be connected to enable the data like that in Table 1 to be collected. **[6 marks]**
- (b) (i) On the grid provided on page 7, use the data below to plot the characteristic (V_{out} vs V_{in}) for a non-inverting amplifier similar to the one in (a) above.

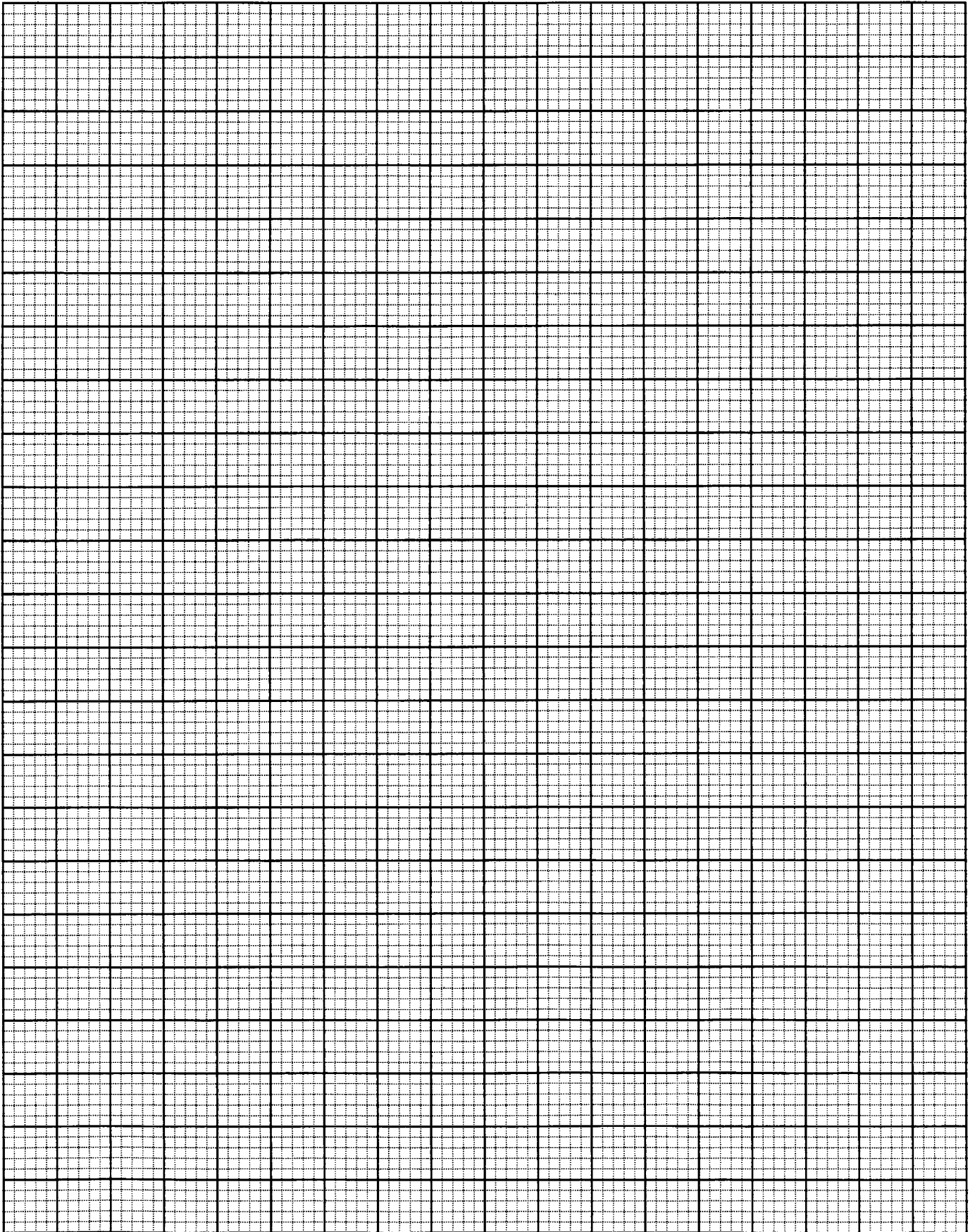
TABLE 1

V_{in} / V	V_{out} / V	V_{in} / V	V_{out} / V
3.00	12.90	0.18	1.74
2.60	12.90	-0.59	-5.82
2.30	12.90	-0.98	-9.72
1.90	12.90	-1.18	-11.70
1.22	12.10	-1.60	-13.80
1.05	10.40	-1.90	-13.80
0.73	7.27	-2.60	-13.80
0.56	5.50	-3.30	-13.80

- (ii) From the graph, determine, to 3 significant figures, the gain of the amplifier. **[9 marks]**

Total 15 marks

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NOTHING HAS BEEN OMITTED.

3. (a) (i) State the name of the scientist who is credited with the explanation of the photoelectric effect.

- (ii) With reference to the photoelectric effect, explain what is meant by

a) 'threshold frequency'

b) 'work function'.

[3 marks]

- (b) Figure 4 shows a circuit used in an experiment to investigate the photoelectric effect from a metal. Table 2 shows data obtained from such an experiment.

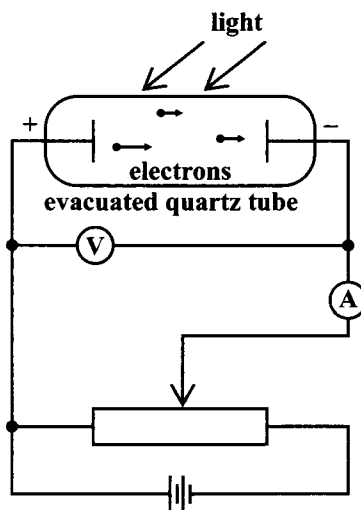


Figure 4

TABLE 2

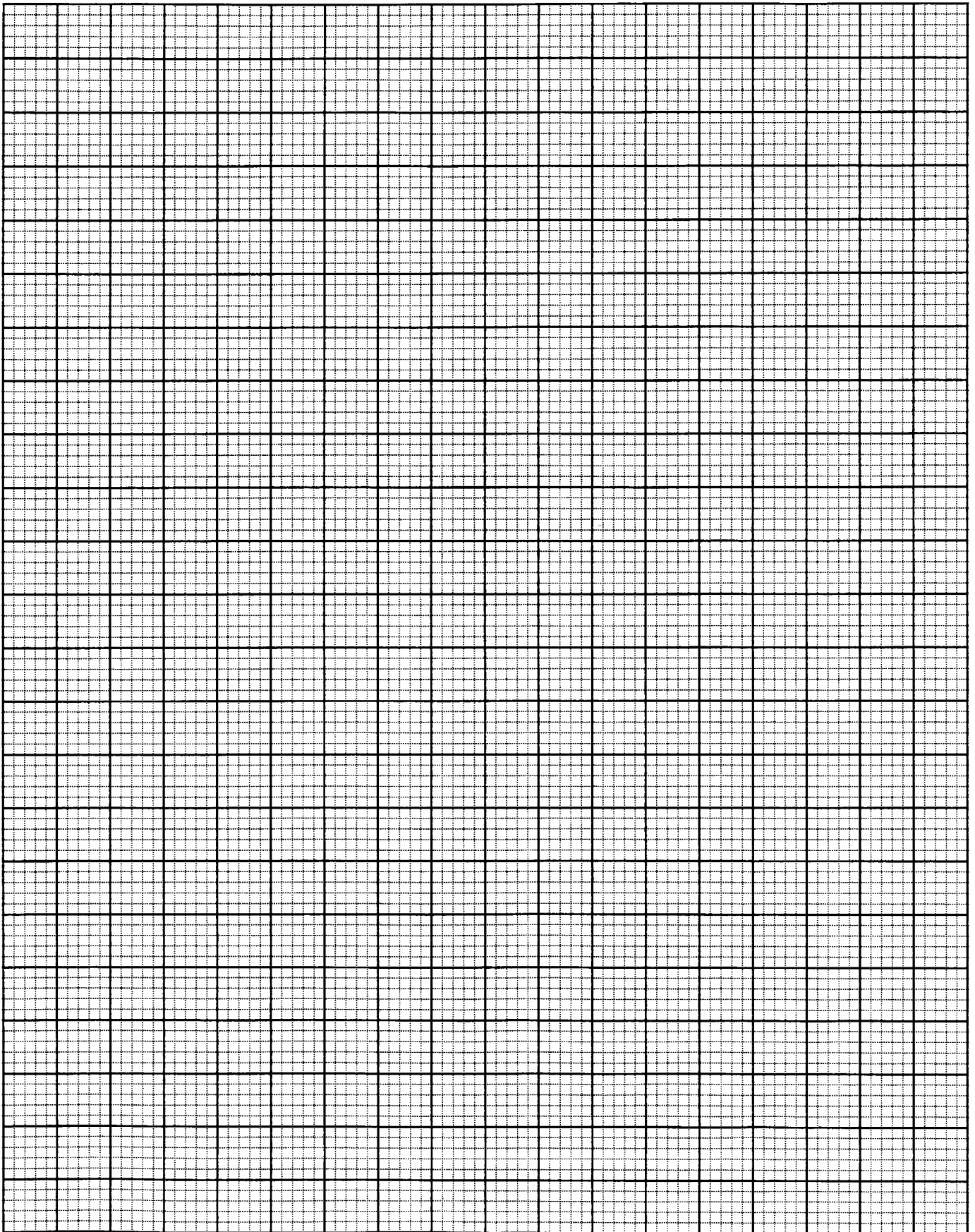
Wavelength λ/nm	Frequency, $f \times 10^{14} \text{ Hz}$	Stopping Potential V_s/volt
544	5.51	0.360
594		0.199
604	4.97	0.156
612		0.117
633	4.74	0.062

- (i) Fill in the missing values in Table 2. **[1 mark]**
- (ii) On the grid provided on page 11, plot a graph of stopping potential energy versus frequency for the metal. **[4 marks]**
- (iii) Use your graph to determine the values given by this data for
- a) Planck's constant
- [4 marks]**
- b) the work function of the metal.

[3 marks]

Total 15 marks

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SECTION B

Answer ALL questions in this section.

Write your answers in the spaces provided at the end of each question.

4. (a) (i) Define the terms 'capacitance' and 'farad'.
- (ii) Write an expression for the net capacitance of three capacitors of capacitance C_1 , C_2 , C_3 connected in
- a) series
- b) parallel.
- (iii) State what is meant by the term 'time constant' for a capacitor discharge circuit. [5 marks]
- (b) Figure 5 shows a circuit diagram for a highway emergency flasher that uses a 120 V battery, a $1.0 \text{ M}\Omega$ resistor, a $1.0 \text{ }\mu\text{F}$ capacitor and a neon flash lamp. The flash lamp has a resistance of more than $1 \times 10^7 \text{ }\Omega$ when the voltage across it is less than 110 V. Above 110 V, the neon gas ionizes, the resistance of the lamp drops to $10 \text{ }\Omega$ and the capacitor then discharges completely in **three time constants** of the discharge circuit.

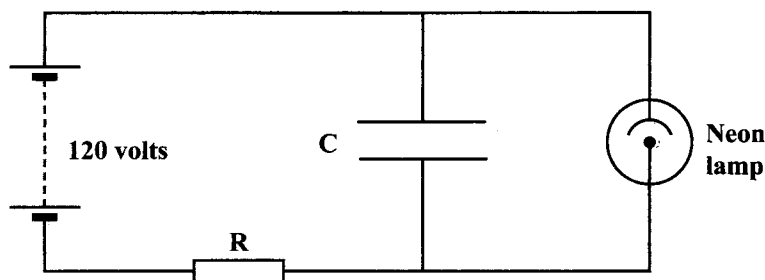


Figure 5

- (i) Calculate EACH of the following:
- a) The time between flashes, that is, the time it takes for the uncharged capacitor to reach a p.d. of 110 V
- [The formula for the charging of a capacitor is $V = V_0 (1 - e^{-\frac{t}{RC}})$]
- b) The duration of each flash
- c) The energy released in each flash [6 marks]
- (ii) It is claimed that during the flash, the light is as bright as that from a 200 W light bulb. Comment on this statement. [4 marks]

Total 15 marks

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Figure 1

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5. Figure 6 shows the variation with frequency, f , of the voltage gain, G , without feedback, of an operational amplifier (op-amp).

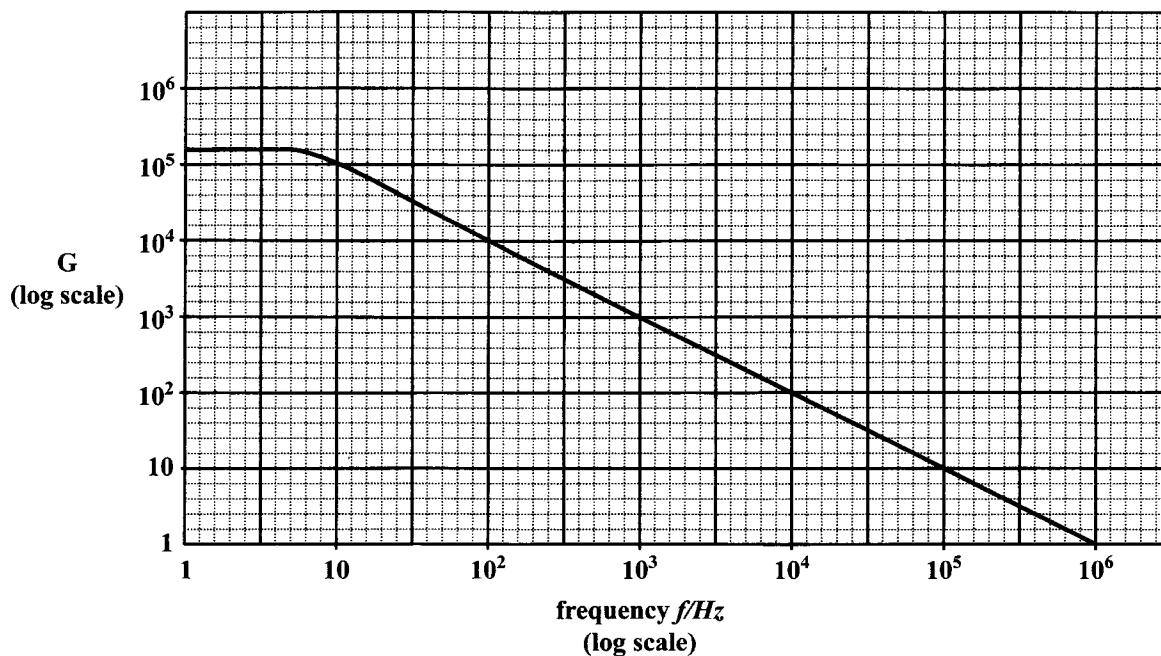


Figure 6

The op-amp is used in the circuit shown in Figure 7.

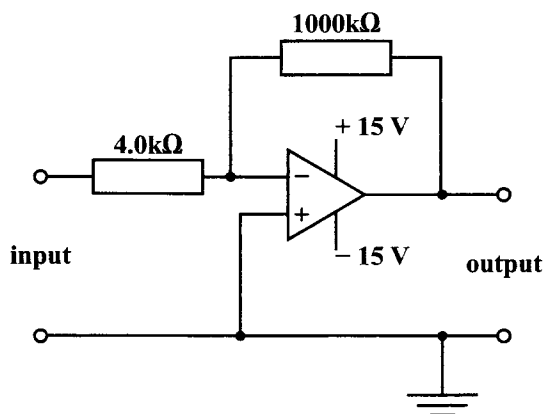


Figure 7

(a) State

- the type of amplifier circuit shown in Figure 7
- what is meant by 'negative feedback' in this amplifier.

[2 marks]

- (iii) Calculate
- the gain of this amplifier at low frequencies
 - the bandwidth of the amplifier. **[5 marks]**
- (iv) What is the peak output voltage for an input signal of peak value 0.052 V and frequency 1.0×10^4 Hz? **[2 marks]**

- (b) Figure 8 shows a digital circuit with inputs A and B.

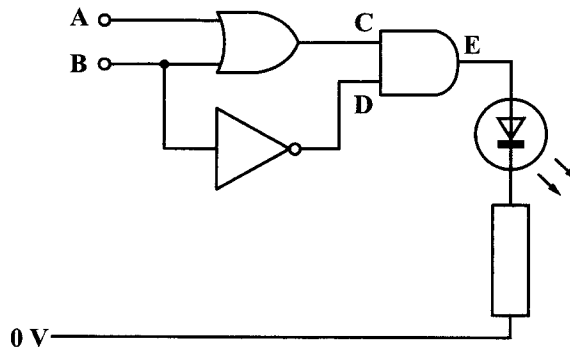


Figure 8

- (i) Copy and complete the truth table (Table 3) for the circuit shown in Figure 8.

TABLE 3

A	B	C	D	E
0	0			
0	1			
1	0			
1	1			

[3 marks]

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- (ii) The circuit of Figure 8 is modified so that inputs A and B are controlled by a thermistor, R_T , and an LDR, R_L , as shown in Figure 9. The thermistor has a negative temperature coefficient and the resistance of R_L is high when in darkness and low when in light.

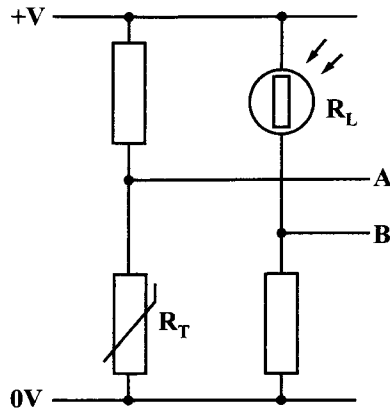


Figure 9

- a) The logic state at A changes when R_T is cooled. State whether R_T is hot or cold for logic 1 at A.
- b) The logic state at B changes when R_L is put in darkness. State whether R_L is in light or darkness for logic 1 at B.
- c) Hence, deduce the environmental conditions for the LED to be conducting.
[3 marks]

Total 15 marks

Write your answer to Question 5 here.

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6. (a) (i) Radon has the nuclear formula $^{226}_{86}\text{Rn}$. State the number of protons, neutrons and electrons that are in the neutral atom.
- (ii) Explain what is meant by the term 'isotope'. [4 marks]
- (b) Define the terms 'half-life' and 'decay constant, λ ' for a radioactive nucleus. Write an equation relating the two quantities. [3 marks]
- (c) A radioactive sample contains $3.50\ \mu\text{g}$ of pure $^{11}_6\text{C}$, which has a half-life of 20.4 minutes. Determine the
- (i) number of nuclei in the sample at $t = 0$
- (ii) decay constant of the sample
- (iii) activity of the sample at $t = 0$
- (iv) activity of the carbon-11 after 8 hours. [8 marks]

Total 15 marks

Write your answer to Question 6 here.

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IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.